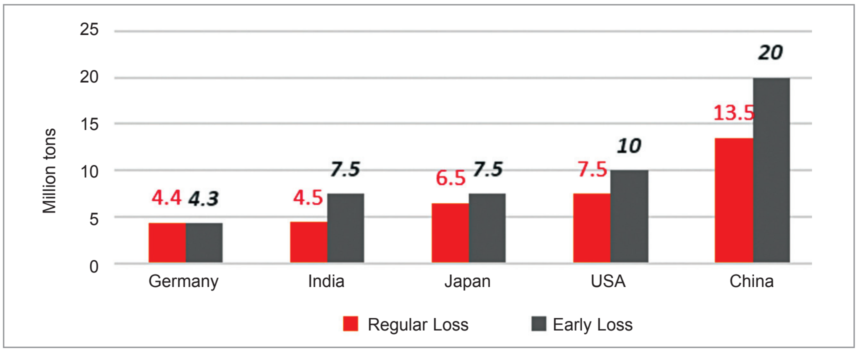

Fig. 5: Projections of cumulated solar PV waste in top 5 countries

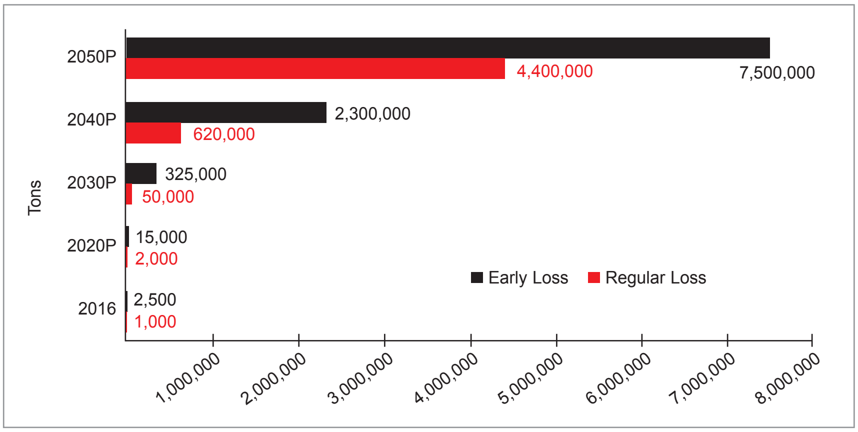

Fig. 6: Projected cumulated solar PV waste in India

waste generation is analysed under two scenarios:

1. Regular loss after 25-30 years
2. Early loss due to qualitative issues and re-powering

Under regular loss, all manufacturers stand a guarantee of performance over a period of 25 years with 90% output for first 12 years and up to 80% after 25 years of operation. At the end of 25 years majority of the modules are discarded as they reach the end of their life as waste. In fact, most of them are becoming waste after 20 years itself.

The causes of early loss can be from the following:

- Glass breakage
- J-box and cables
- Power loss
- Delamination
- Transport damage
- Loose frame
- Optical failure
- Defect cell interconnect
- Re-powering with high efficiency cells

As per study undertaken by International Renewable Energy Agency (IRENA) – International Energy Agency (IEA) Photovoltaic Power System (PVPS) programme, the worldwide solar module waste would grow to 78 million metric tons by 2050. The top five countries by themselves will account for about 50 million metric tons as shown in Fig. 5.

It is interesting to note that India will have the highest early loss and their causes need to be investigated systematically.

Using Fig. 1, the accumulated solar PV waste in India—considering a standard 25-year life under regular loss—can be estimated. However, going by the projections in Fig. 6, it is difficult to estimate the exact volume of waste as it is anticipated many modules may achieve early loss. This thought needs to be examined with specific case to India by undertaking a survey amongst manufacturers. The IRENA-IEA PVPS modelling provides insight on the cumulative solar PV waste under regular and early loss scenarios as shown in Fig. 6.

If these PV module wastes are left lying under the early loss scenario then considering standard 12.2-metre (40 feet) container the distance occupied will be as below in Table 2.

Material-wise projection of waste in India

Table 3 shows the recovered quantity of materials from solar PV waste in India under early-loss scenario post their end of life.

International regulations

At present, only the European Union (EU) has adopted and pioneered PV-specific waste regulations. Most countries around the world classify PV panels as general or industrial waste. In limited cases, such as in Japan or the US, general waste regulations may include panel testing for hazardous material content as well as prescription or prohibition of specific shipment, treatment, recycling, and disposal pathways.

Estimation of Area Occupied by PV Module Waste in India Under Early Loss Scenario — Table 2

Year	PV waste early loss (tons)	Tons/MW	Capacity per container (MW)	Number of containers	Length per 40ft container (m)	Distance occupied (km)	
2016	2,500	100	0.25	100	12.20	1	
2020P	15,000	85	0.25	706	12.20	9	Mumbai-Dadar
2030P	325,000	70	0.25	18,571	12.20	227	Delhi-Agra
2040P	2,300,000	65	0.25	141,538	12.20	1,727	Delhi-Chennai
2050P	7,500,000	60	0.25	500,000	12.20	6,100	Delhi-Ahmedabad-Mumbai-Bengaluru-Chennai-Kolkata-Delhi